Pointless Coherence

A short kit: when a coherent whole has no points, not even ideal ones

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The question. An earlier insight (CE) was that coherence does not force σ-completion. On a Boolean algebra a coherent two-valued state is still a *point*, though possibly an ideal one (an ultrafilter, by Stone). The horizon was open, but it was made of points. This kit asks whether a coherent whole can have *no* point, not even an ideal one. It also asks what the cycle and the completed infinite each contribute.

Path.

- Problem 1: pointlessness alone is cheap. It happens on six points, and there it costs nothing at the σ level.
- Problem 2: on the glued ring, demanding σ-additivity *forces* a point.
- Problem 3: coherence alone does not. Coherent ideal points sit on every window, with no ideal point for the whole.
- Problem 4: cut the cycle and ideal points always glue.
- Problem 5: what is still open.

Status of what is here.

- Problems 1–4 are elementary. The proofs in Part III are mine and have not been independently checked.
- Problem 4 is very likely folklore (amalgamation of ultrafilters, or of types, over trees).
- The step from window level to the whole carrier is a Lean-checked fact, quoted and not rebuilt here. Rebuilding it is the hard part.
- Problem 5 is a conjecture.

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Part I
Problems

Shared definitions

Definition 1 (carrier, state, point)**.** A *carrier* is a family $\mathcal{D}$ of subsets of a set Ω containing Ω and closed under complements and countable disjoint unions (a Dynkin system, or concrete σ-orthoposet). A *two-valued state* is $s : \mathcal{D} \to \{0, 1\}$ with $s(\Omega) = 1$ and $s(A \sqcup B) = s(A) + s(B)$ for disjoint $A, B \in \mathcal{D}$ (finite additivity). It is *σ-additive* if this holds for countable disjoint unions. It is a *point* if $s = \delta_\omega$, that is $s(E) = [\omega \in E]$, for some $\omega \in \Omega$.

Definition 2 (ultrafilter; ideal point)**.** A non-principal ultrafilter p on $\mathbb{N}$ is a $\{0, 1\}$-valued finitely additive measure on all of $\mathcal{P}(\mathbb{N})$ that vanishes on finite sets. It contains every cofinite set. An *ideal point* of Ω is an ultrafilter W on Ω. A state s on $\mathcal{D}$ *is an ideal point* if $s(E) = [E \in W]$ for some ultrafilter W on Ω. Points are the principal ideal points.

Definition 3 (the glued ring)**.** Fix $L \geq 3$ and $\Omega = \mathbb{N}^L$, indices mod L. The *window* $W_k = \{k, k{+}1\}$. A *window event* at k is a set $\{x : (x_k, x_{k+1}) \in A\}$ with $A \subseteq \mathbb{N}^2$ arbitrary. The *glued carrier* $\mathcal{D}_L$ is the smallest carrier containing all window events. Set $C_k = \{x : x_k < x_{k+1}\}$, a window event at k.

Definition 4 (ordered product)**.** For ultrafilters p, q on $\mathbb{N}$, the ultrafilter $p \otimes q$ on $\mathbb{N}^2$ is

$$A \in p \otimes q \iff \{m : \{n : (m, n) \in A\} \in q\} \in p.$$

Read it as "first draw m by p, then draw n by q, with n chosen *after* m".

1 The problems

Problem 1 (pointless, but finite)**.** *Let* $\Omega = \{1, \ldots, 6\}$*,* $M_1 = \{1, 2, 3\}$*,* $M_2 = \{3, 4, 5\}$*,* $M_3 = \{5, 6, 1\}$ *and* $M_4 = \{2, 4, 6\}$*.*

(a) Show that the carrier generated by the M_i is exactly $\{\emptyset, \Omega, M_i, M_i^c\}$.

(b) Show that $s(M_i) = 0$ for all i defines a two-valued state.

(c) Show that s is not a point.

(d) Explain why s is nevertheless σ-additive, so that the σ-completion asks nothing of it.

Source: Navara–Pták, Čas. pěst. mat. 108 (1983); on the shelf.

Why it matters. Pointlessness on its own is cheap. What is new later is pointlessness that the σ-completion *forbids*. Note the cyclic overlaps: M_1, M_2, M_3 meet in a ring at $3, 5, 1$.

Problem 2 (σ-additivity forces a point)**.** *Let s be a σ-additive two-valued state on $\mathcal{D}_L$. Show that $s = \delta_\omega$ for some $\omega \in \mathbb{N}^L$. Deduce that no such s has $s(C_k) = 1$ for every k.*

Source: Hub `collision_vs_generation.md`, "Reduction"; Lean `WindowGluing`, `no_sigma_extension`.

Why it matters. On this carrier, the completed infinite is *the demand to be made of points*. The pattern "every step goes up" is then impossible, by cyclic order.

Problem 3 (the Escher staircase: ideal points on every window, none for the whole)**.** *Let p be a non-principal ultrafilter on $\mathbb{N}$ and $L \geq 3$. On window k put $u_k = p \otimes p$, read on coordinates (x_k, x_{k+1}).*

(a) *Show that* $\{(m,n) : m < n\} \in p \otimes p$. *So every window says "up".*

(b) *Show that both marginals of* $p \otimes p$ *are* p. *Deduce that the window ultrafilters agree on shared coordinates.*

(c) *Show that no ultrafilter* W *on* $\mathbb{N}^L$ *projects to every* u_k. *Stronger: no two-valued finitely additive state on the* algebra *generated by the window events agrees with every* u_k.

(d) *Show that* $p \otimes p \neq$ *its transpose. Where does the proof of (a) use the order of the draws?*

Quoted, not to rebuild: *the window states extend to a two-valued finitely additive state on the whole carrier* $\mathcal{D}_L$, *for every* $L \geq 4$ *(Lean,* `mu_cases` *and* `sL_sigmaEssential`*). The extension is therefore coherent, it is not an ideal point (by (c)), and by Problem 2 it has no* σ*-additive version.*

Source: Hub `collision_vs_generation.md`, "The pattern"; Lean `targetA_sharp_countable_literal`.

Why it matters. This is a coherent whole with no substrate, not even an ideal one. Locally, every coordinate *is* an ideal point (p). The non-commutativity of $\otimes$ gives each window an arrow, and the cycle closes the arrow on itself.

Problem 4 (cut the cycle and ideal points glue)**.** *(a) (*Amalgamation.*) Let* U *be an ultrafilter on* $X \times Y$ *and* V *one on* $Y \times Z$, *with the same* Y*-marginal. Show there is an ultrafilter on* $X \times Y \times Z$ *whose projections are* U *and* V.

(b) *(*Paths.*) On* $\mathbb{N}^{n+1}$ *with windows* $\{k, k{+}1\}$ *for* $0 \leq k < n$ *(a path, not a ring), show that any ultrafilters* u_k *on the windows that agree on shared coordinates are the projections of one ultrafilter on* $\mathbb{N}^{n+1}$.

(c) *Apply (b) to the staircase of Problem 3 with one window removed. Which global ultrafilter do you get, and what order does it put on the coordinates?*

Why it matters. At the level of windows, this is a Vorob'ev theorem for ideal points. On an acyclic overlap structure, coherent ideal points always glue. The cycle is what breaks them, and Problem 3 shows it can.

Problem 5 (open: the whole carrier)**.** *On an acyclic glued carrier (a path of windows, the Dynkin closure of its window events), is every two-valued finitely additive state an ideal point?*

Problem 4 gives an ultrafilter W *that agrees with* s *on window events. It does* not *show that* W *agrees with* s *on all of* $\mathcal{D}$. *A finitely additive state is not determined by its values on generators, because countable disjoint unions in* $\mathcal{D}$ *are not controlled. That gap is exactly where the Lean proof* `mu_cases` *does its work on the ring.*

Why it matters. A yes answer would give the full statement: *acyclic implies coherent two-valued states are ideal points; a cycle permits pointless coherence.* That would be one theorem uniting the cyclic marginal problem and the σ gap. Prior art to check first: ultrafilter and type amalgamation (Goldberg, arXiv:2102.04677), and Bhaskara Rao, *Theory of Charges*.

Things to sit with (not exercises)

- **The finite example.** In Problem 1 the whole has no point but needs none: finiteness makes it σ-additive for free. In Problem 3 the σ-completion is *not* a neutral scaffold. It legislates points, and in doing so excludes something coherent.
- **Where the ideal element sits.** The anti-Platonic object (a whole without points) is built from a Platonic element (a non-principal ultrafilter, which needs choice) placed *locally*. Is that an accident of the construction, or necessary?

- **The arrow.** $p \otimes p$ carries an order: "drawn later" means "larger". Each window carries a local time. The ring asks for a global time, and there is none. Compare recurrence as cyclic time in the flagship.

Part II
Hints

1. (a) Every member listed has three elements, or none, or six. Which pairs are disjoint? (c) A point ω would lie outside every M_i. Compute $\bigcap M_i^c$.

2. The window events at k form a copy of $\mathcal{P}(\mathbb{N}^2)$, a Boolean σ-algebra on a countable set. Use the singletons there. Then: the set of $E \in \mathcal{D}_L$ on which two σ-additive states agree is closed under complements and countable disjoint unions.

3. (a) For fixed m, $\{n : n > m\}$ is cofinite. (c) An ultrafilter, or a two-valued charge on an algebra, is closed under finite intersections. What is $\bigcap_k C_k$? (d) Apply (a) to the transposed set $\{(m, n) : m > n\}$.

4. (a) Show that $\{A \times Z : A \in U\} \cup \{X \times B : B \in V\}$ has the finite intersection property, using $\pi_Y(A) \in q$. Extend it to an ultrafilter, and use maximality for the projections. (b) Induct along the path.

Part III
Proofs

Written by Claude; not independently checked.

Problem 1.

(a) The family $\mathcal{F} = \{\emptyset, \Omega, M_i, M_i^c\}$ is closed under complements. Its proper members all have three elements. Two disjoint three-element subsets of a six-element set are complements of each other. So the only disjoint unions of proper members are $M \sqcup M^c = \Omega$, and unions with $\emptyset$ are trivial. Hence $\mathcal{F}$ is a carrier. It is the smallest one containing the M_i, since any such carrier contains their complements.

(b) Put $s(M_i) = 0$, $s(M_i^c) = 1$, $s(\emptyset) = 0$, $s(\Omega) = 1$. Every disjoint pair is $\{E, E^c\}$ or contains $\emptyset$, and the values add correctly in each case.

(c) $M_1^c \cap M_2^c \cap M_3^c \cap M_4^c = \{4, 5, 6\} \cap \{6, 1, 2\} \cap \{2, 3, 4\} \cap \{1, 3, 5\} = \emptyset$, so no ω has $\delta_\omega(M_i) = 0$ for all i.

(d) A countable disjoint family in a finite carrier has only finitely many non-empty members. So finite additivity is σ-additivity.

Problem 2.

- *Each window gives a point.* The window events at k are the preimages of $\mathcal{P}(\mathbb{N}^2)$ under $x \mapsto (x_k, x_{k+1})$. The singletons $\{(a, b)\}$ partition $\mathbb{N}^2$ countably. By σ-additivity, exactly one fibre $\{x : (x_k, x_{k+1}) = (a_k, b_k)\}$ has value 1. Monotonicity then gives $s(E) = [(a_k, b_k) \in A]$ for every window event E at k, with base set A.

- *The points agree.* Windows k and $k{+}1$ share coordinate $k{+}1$. The event $\{x_{k+1} = c\}$ is a window event of both, so $b_k = a_{k+1}$. Define ω by $\omega_k = a_k$. Then $s = \delta_\omega$ on all window events.
- *Extend to all of $\mathcal{D}_L$.* The class $\{E \in \mathcal{D}_L : s(E) = \delta_\omega(E)\}$ contains Ω and the generators. Since both states are σ-additive, it is closed under complements and countable disjoint unions, so it is all of $\mathcal{D}_L$.
- *The pattern fails.* If $s(C_k) = 1$ for all k, then $\omega_0 < \omega_1 < \cdots < \omega_{L-1} < \omega_0$, which is impossible.

Problem 3.

(a) For each m, the section $\{n : n > m\}$ is cofinite, so it lies in p. The set of such m is $\mathbb{N} \in p$.

(b) First marginal: $\{(m, n) : m \in X\} \in p \otimes p$ iff $\{m \in X\} \in p$, because each section is $\mathbb{N}$ or $\emptyset$. Second marginal: every section of $\{(m, n) : n \in Y\}$ is Y, so the set lies in $p \otimes p$ iff $Y \in p$. Window k's second coordinate and window $k{+}1$'s first coordinate are both x_{k+1}, and both carry p.

(c) An ultrafilter projecting to every u_k would contain every C_k, hence $\bigcap_k C_k = \emptyset$, which is absurd. The same holds for any two-valued finitely additive state on the algebra generated by the windows. Such a state is multiplicative ($s(A \cap B) = 1$ when $s(A) = s(B) = 1$, because $A \cap B^c$ and B^c give $s(A \cap B^c) = 0$), and the algebra contains $\bigcap_k C_k$.

(d) By the argument of (a), $\{(m, n) : m < n\} \in p \otimes p$. Its transpose, the image of that set under swapping coordinates, is $\{(m, n) : m > n\}$, which is disjoint from it. If $p \otimes p$ equalled its transpose, both disjoint sets would belong to it, which is absurd. Proof (a) used that the *inner* variable n is drawn by a non-principal ultrafilter *after* m is fixed.

Problem 4.

(a) Let q be the common Y-marginal. Both U and V are closed under finite intersections, so it suffices to show $(A \times Z) \cap (X \times B) \neq \emptyset$ for $A \in U$ and $B \in V$. Now $A \subseteq \pi_Y^{-1}(\pi_Y A)$ gives $\pi_Y(A) \in q$, and likewise $\pi_Y(B) \in q$. So some y lies in both. Pick $(x, y) \in A$ and $(y, z) \in B$; then (x, y, z) is in the intersection. Extend to an ultrafilter W. Its $X \times Y$ projection is an ultrafilter containing U, hence equal to U; likewise for V.

(b) Amalgamate u_0, u_1 on coordinates $\{0, 1, 2\}$. Then amalgamate the result with u_2 along coordinate 2, and so on, using (a) with $X = \mathbb{N}^{\{0..k\}}$, $Y = \mathbb{N}$ and $Z = \mathbb{N}$.

(c) With windows $0, \ldots, L{-}2$ kept, any amalgam W contains $C_0 \cap \cdots \cap C_{L-2} = \{x_0 < x_1 < \cdots < x_{L-1}\}$. That is an ascending staircase with nothing closing it, and it is fine. The removed window would ask $x_{L-1} < x_0$.

Problem 5. Open. See the statement.